

A. Additional Details for Point Cloud Reconstruction

For each observation, DEIL acquires RGB images I_i , instance masks M_i , camera intrinsics K_i , and extrinsics E_i . In simulation, the masks and camera parameters are directly retrieved from the simulator. In real-world experiments, object masks are initialised and tracked using Segment-and-Track-Anything [1] framework. The camera intrinsics and extrinsics are obtained through calibration.

1) *Depth prediction and segmentation*: The RGB images from all selected views are passed to the chosen depth estimator. In the masked-RGB condition, pixels outside the RL Bench object mask are set to zero before inference. Importantly, the object mask is also applied during point-cloud reconstruction for both the RGB and masked-RGB conditions; thus, masking the estimator input and segmenting the final geometry are separate operations. To reduce mixed-depth artifacts at object boundaries, each mask is eroded once with a square kernel of size 2×2 , 3×3 , or 5×5 for resolutions 128, 256, and 512, respectively. Although some depth estimation models also predict point maps and camera parameters, DEIL uses only their predicted per-view z -depth maps. This provides a common reconstruction procedure across all methods: the predicted depth is combined with the same intrinsics and extrinsics. The ground-truth baseline in simulator uses RL Bench metric depth directly and otherwise follows the same segmentation and back-projection procedure.

2) *Depth-estimator settings*: We use the default settings for each of the depth estimator, only modify slightly to suit each of our experiment configurations.

- π^3 [2]. Images are resized so that their total area is at most 255,000 pixels, with both dimensions constrained to multiples of 14. Inference uses bfloat16 on supported GPUs and float16 otherwise. The z -coordinate of the predicted local point map is used as depth and resized to the original resolution using bilinear interpolation.
- VGGT [3]. Views are processed jointly using mixed precision. Images are resized to a width of 518 pixels while preserving aspect ratio and centre-cropped where necessary; predicted depth is resized to the original resolution using nearest-neighbour interpolation. Predicted confidence, point maps, and camera parameters are not used in the final reconstruction.
- MoGe [4]. Each view is processed independently after scaling RGB values to $[0, 1]$. Negative depths and depths greater than 10 m are set to zero.
- MAST3R [5]. Images are loaded at size 512, all image pairs are included in a complete symmetrised scene graph, and sparse global alignment is run with shared intrinsics and depth optimisation. The default coarse and refinement stages use 300 iterations each. Cleaned dense depth is resized to the observation resolution using bilinear interpolation.
- MapAnything [6]. For the calibrated configuration, RL-Bench intrinsics and camera poses are converted to

the OpenCV camera convention and supplied as metric calibration. Inference uses bfloat16 mixed precision, internal geometry masking, and edge masking. The predicted z -depth is then resized bilinearly and clipped.

3) *Back-projection*: For pixel coordinates (u, v) and predicted depth $d_i(u, v)$, the camera-frame point is

$$\mathbf{p}^{C_i}(u, v) = \begin{bmatrix} (u - c_{x,i})d_i(u, v)/f_{x,i} \\ (v - c_{y,i})d_i(u, v)/f_{y,i} \\ d_i(u, v) \end{bmatrix}.$$

The corresponding world-frame point is

$$\tilde{\mathbf{p}}^W(u, v) = E_i \begin{bmatrix} \mathbf{p}^{C_i}(u, v) \\ 1 \end{bmatrix}.$$

Non-finite and negative depths are set to zero before projection. Only pixels retained by the eroded object mask contribute to the cloud.

4) *Multi-view scale alignment and fusion*: Learned depth estimations from different views can have inconsistent scales. For multi-view observations, DEIL therefore estimates one positive scalar s_i per view while keeping the calibrated camera poses fixed. Each point cloud is scaled about its camera centre $\mathbf{t}_i$:

$$\hat{P}_i(s_i) = \{\mathbf{t}_i + s_i(\mathbf{p} - \mathbf{t}_i) \mid \mathbf{p} \in P_i\}.$$

Scaling about the camera centre preserves the predicted viewing rays. The scales are constrained to $[0.01, 20]$ and estimated using a symmetric 80%-trimmed Chamfer objective between adjacent views:

$$\mathcal{L}(\mathbf{s}) = \frac{1}{2(V-1)} \sum_{i=1}^{V-1} \left[D_{0.8}(\hat{P}_i, \hat{P}_{i+1}) + D_{0.8}(\hat{P}_{i+1}, \hat{P}_i) \right],$$

where $D_{0.8}(A, B)$ is the mean squared nearest-neighbour distance after retaining the lowest 80% of distances. At most 512 finite points per view are randomly selected for optimisation. Optimisation is performed in log-scale using differential evolution, followed by alternating fixed-correspondence bounded least-squares updates and a final Powell refinement. The tolerance is 10^{-5} . Our pipeline uses $n_{init} = 8$ and at most 30 iterations per stage. Scale alignment is skipped for metric ground-truth depth or single-camera observations. The estimated scales are applied to all retained points, after which the per-view clouds are concatenated.

5) *Policy-specific preprocessing*: Only XYZ coordinates are provided to the policies; RGB point features and workspace cropping are not used. For 3D Diffusion Policy, the fused cloud is reduced to 1,024 points using farthest-point sampling. For Instant Policy, $3N$ candidates are first sampled, Open3D statistical outlier removal is applied with 10 neighbours and a standard-deviation ratio of 3.0, and $N = 2,048$ points are then sampled uniformly. Sampling with replacement is used when insufficient points are available. Finally, the world-frame point cloud is transformed into the end-effector frame. The same representation and preprocessing are used for demonstrations and online policy observations.

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